

KEYS DECIDE, VALUES DESCRIBE: ACCEPTANCE-AWARE KV CACHE COMPRESSION FOR SPECULATIVE DECODING

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ABSTRACT

Speculative decoding accelerates language-model inference by asking a small draft model to propose tokens for verification by a larger target model, but standard implementations retain a separate key-value (KV) cache for each model. We study how the draft cache can be compressed while preserving the quantity that directly controls speculative speed: token acceptance. Our central observation is that acceptance is strongly asymmetric in key and value precision. On a Qwen2.5-3B target with a Qwen2.5-1.5B draft at 4K context, 8-bit keys and 4-bit values reduce total target-plus-draft KV memory by 27.0% while retaining 95.4% of native acceptance (0.544 versus 0.571). Reversing the assignment gives the same memory saving but retains only 79.6% of acceptance (0.454). Controlled Gaussian perturbations expose the same mechanism: key noise changes attention routing and rapidly destroys top-1 stability, whereas value noise is tolerated at much larger magnitudes. Based on this observation, SPECKV profiles each draft layer and cache component under the actual speculative loop, then selects mixed precision under an acceptance-drop budget. Exact full-precision target verification preserves the target output distribution. We evaluate the hypothesis across Qwen2.5, Qwen3, Llama 3, OLMo 2, and SmolLM2 model pairs and across 1K–4K contexts.

1 INTRODUCTION

Speculative decoding converts several serial target-model decoding steps into a single parallel verification step by sampling proposals from a cheaper draft model (Leviathan et al., 2023; Chen et al., 2023). The method is distribution preserving, but its systems footprint includes two model states and two growing KV caches. This duplication becomes consequential at long context, where each generated token reads the prefix cache and cache capacity limits the number of concurrently served requests.

Quantizing the draft cache seems straightforward, but generic model-quality metrics do not reveal whether a compressed cache remains a useful speculator. A small distributional error can move the draft’s top proposal away from the target’s token and terminate an entire speculative block. Conversely, errors in directions irrelevant to the verifier may have little effect on acceptance. This motivates an acceptance-aware rather than reconstruction-aware objective.

Our investigation starts from the attention decomposition

$$A_\ell = \text{softmax}\left(\frac{Q_\ell K_\ell^\top}{\sqrt{d_h}} + M\right), \quad O_\ell = A_\ell V_\ell W_{O,\ell}. \quad (1)$$

Keys determine the routing distribution A_ℓ , whereas values determine the payload read after routing. We hypothesize that coarse key quantization is more likely to redirect attention and cascade through subsequent layers than an equal-memory reduction in value precision. This role asymmetry is compatible with prior KV quantization analyses (Liu et al., 2024; Hariri et al., 2025), but we ask a different operational question: *which errors can a draft model tolerate without losing speculative acceptance?*

Our preliminary findings support three claims. First, Gaussian key perturbations are dramatically more destructive to next-token stability than value perturbations. Second, K8V4 consistently dominates

K4V8 on Qwen2.5 despite their identical memory cost. Third, a conservative layer-wise allocator can reduce the draft cache while holding acceptance close to the full-precision draft. The ongoing cross-family campaign tests whether this acceptance asymmetry generalizes across architecture, scale, and context length.

2 BACKGROUND AND PROBLEM FORMULATION

Let $p(\cdot \mid x_{\leq t})$ be the target distribution and $q(\cdot \mid x_{\leq t}; \hat{\mathcal{C}}_d)$ the draft distribution produced from a compressed draft cache $\hat{\mathcal{C}}_d$. Standard speculative sampling accepts proposals using the target correction rule; greedy speculative decoding accepts the longest proposal prefix matching target argmax tokens. In both cases, target verification makes the final sequence exact with respect to the chosen target decoding rule. Compression affects efficiency through acceptance, not final correctness.

For target and draft cache sizes M_t and M_d , native speculative decoding uses $M_t + M_d$. If layer ℓ stores keys and values with precisions b_ℓ^K, b_ℓ^V , the compressed draft cost is approximately

$$\widehat{M}_d \propto T \sum_{\ell=1}^{L_d} H_\ell^{KV} d_{h,\ell} (b_\ell^K + b_\ell^V), \quad (2)$$

plus one scale per token and KV head in our per-vector quantizer. Total saving is $(M_d - \widehat{M}_d)/(M_t + M_d)$; it is necessarily smaller than the percentage saved from the draft cache alone.

3 METHOD

3.1 DRAFT-CACHE FAKE QUANTIZATION

For a key or value vector x , we use symmetric per-vector fake quantization,

$$s(x, b) = \frac{\max_i |x_i|}{2^{b-1} - 1}, \quad \hat{x} = s \cdot \text{clip}(\text{round}(x/s), -2^{b-1} + 1, 2^{b-1} - 1). \quad (3)$$

Only the draft KV state is quantized between cached decode steps. The target model and verifier remain in BF16. This simulator measures the algorithmic memory–acceptance tradeoff without conflating it with a particular packed-cache kernel.

3.2 ACCEPTANCE-AWARE SENSITIVITY PROFILES

We first evaluate a full-precision draft baseline. Then, for each selected layer ℓ , component $c \in \{K, V\}$, and candidate bit width $b \in \{4, 8\}$, we quantize only (ℓ, c) while leaving all other draft-cache entries in BF16. The measured risk is

$$r_{\ell,c,b} = \max(0, \alpha_{\text{BF16}} - \text{alpha}_{\ell,c,b}), \quad (4)$$

where α is the end-to-end speculative acceptance rate. We additionally log accepted tokens per verification round, top-1 match, acceptance mass, Jensen–Shannon divergence, and exact agreement with greedy target decoding.

3.3 BUDGETED MIXED-PRECISION ALLOCATION

Given one-at-a-time risks, our transparent initial allocator solves an additive surrogate:

$$\min_{\{b_\ell^c\}} \widehat{M}_d \quad \text{s.t.} \quad \sum_{\ell,c} r_{\ell,c,b_\ell^c} \leq \epsilon, \quad b_\ell^c \in \{4, 8, 16\}. \quad (5)$$

The additive approximation is deliberately conservative and is validated by running the selected allocation through the complete speculative loop. A stronger version will model cross-layer interactions rather than assuming independent risks.

Table 1: Cross-family campaign. Each pair is evaluated at 1K and 4K context, followed by top-eight-layer K/V sensitivity profiling and an acceptance-budgeted allocation.

Family	Target	Draft
Qwen2.5	3B, 7B	1.5B, 3B
Qwen3	8B	4B
Llama 3	Llama-3.1-8B	Llama-3.2-3B
OLMo 2	7B	1B
SmolLM2	1.7B	360M

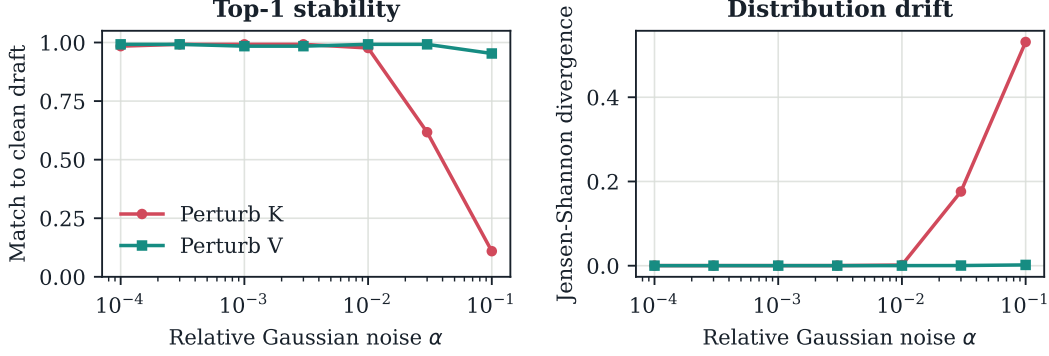


Figure 1: Controlled perturbations of the Qwen2.5-1.5B draft cache. Keys are far more sensitive than values under both top-1 stability and distribution drift.

4 EXPERIMENTAL SETUP

Our preliminary experiment uses Qwen2.5-3B as target and Qwen2.5-1.5B as draft (Yang et al., 2024). We evaluate contiguous WikiText-2 validation blocks, four draft proposals per verification round, greedy target verification, and 16 generated tokens per prompt. The 1K experiment uses 64 prompts; the 4K experiment uses 16 prompts. The ongoing cross-family campaign uses the pairs in Table 1. All target and draft pairs were checked for tokenizer compatibility before launch.

We report acceptance rate, accepted tokens per target verification, top-1 match, JS divergence, acceptance mass, draft-cache bytes, and total target-plus-draft KV bytes. Python fake quantization does not provide a valid kernel-level speed measurement, so we do not make throughput claims from this implementation.

5 PRELIMINARY RESULTS

5.1 KEYS ARE THE FRAGILE ROUTING STATE

Figure 1 perturbs all draft-cache keys or values with Gaussian noise scaled to the clean activation standard deviation. At $\alpha = 0.03$, key noise reduces top-1 agreement with the clean draft to 0.617 and raises JS divergence to 0.176. Value noise at the same magnitude retains 0.992 top-1 match with JS divergence 0.0004. At $\alpha = 0.1$, the corresponding top-1 matches are 0.109 for K and 0.953 for V. This large separation is consistent with routing errors in K changing A in Equation 1 before values are read.

5.2 EQUAL MEMORY DOES NOT IMPLY EQUAL ACCEPTANCE

Figure 2 isolates the core comparison. K8V4 and K4V8 have identical average precision and save the same KV bytes. At 1K context, K8V4 yields 0.475 acceptance versus 0.292 for K4V8. At 4K, the rates are 0.544 and 0.454, respectively. The 4K K8V4 configuration retains 95.4% of native acceptance while saving 61.7% of draft KV memory and 27.0% of combined target-plus-draft KV memory.

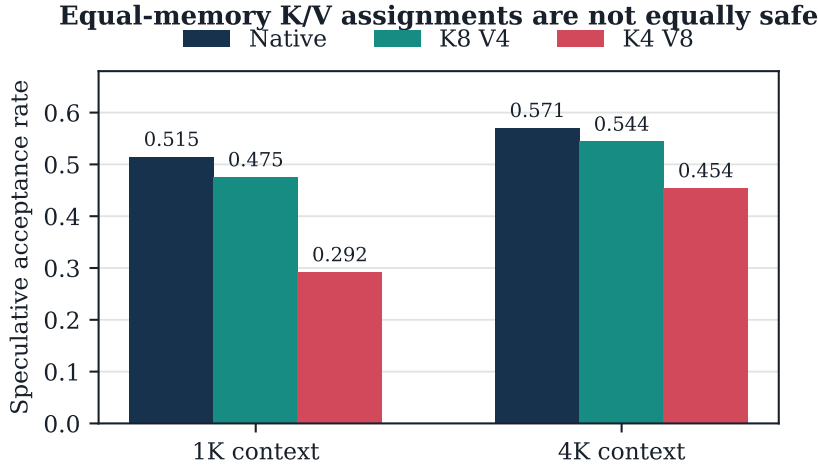


Figure 2: Speculative acceptance for two equal-memory asymmetric assignments. Preserving key precision consistently dominates preserving value precision.

Table 2: Preliminary Qwen2.5-3B/1.5B results. “Retained” normalizes acceptance by the full-precision draft at the same context. Memory includes 16-bit scale metadata for each quantized KV vector.

Context	Draft KV	Accept.	Retained (%)	Draft saved (%)	Total saved (%)	JS ↓
1K	FP16	0.515	100.0	0.0	0.0	0.071
1K	K8 V8	0.495	96.1	49.2	21.5	0.086
1K	K8 V4	0.475	92.4	61.7	27.0	0.088
1K	K4 V8	0.292	56.7	61.7	27.0	0.189
1K	K4 V4	0.300	58.3	74.2	32.5	0.189
4K	FP16	0.571	100.0	0.0	0.0	0.051
4K	K8 V8	0.552	96.8	49.2	21.5	0.068
4K	K8 V4	0.544	95.4	61.7	27.0	0.069
4K	K4 V8	0.454	79.6	61.7	27.0	0.142
4K	K4 V4	0.408	71.5	74.2	32.5	0.160

5.3 A CONSERVATIVE LEARNED ALLOCATION

Profiling the top eight draft layers and selecting precision under a five-point additive acceptance budget gives a deliberately conservative allocation. On 64 held prompts at 1K context, acceptance changes from 0.515 to 0.507 while saving 15.0% of draft-cache memory and 6.56% of total KV memory. This validates the profiling signal but leaves substantial room for a joint, interaction-aware allocator.

5.4 MOTIVATING NEGATIVE RESULT: DIRECT CROSS-MODEL CACHE MAPS

Our earlier approach attempted to replace parts of the draft cache with linear maps from target-model KV states. Sharing the top four of 28 draft layers reduced total KV memory by only 6.25% and reduced acceptance from 0.497 to 0.455 in the cached-quality benchmark. Sharing every layer degraded acceptance much more. This failure motivated the current direction: preserve each model’s native cache semantics, compress the cheaper draft state, and optimize the quantity that matters for speculative decoding. Concurrent work on cross-model prefill reuse also finds that transfer quality depends strongly on error placement and often benefits from RoPE removal, cross-layer selection, or nonlinear maps (Heo et al., 2026).

6 RELATED WORK

KIVI establishes that key and value caches have different outlier structures and uses per-channel K and per-token V quantization (Liu et al., 2024). KV-AdaQuant argues for allocating more precision to keys based on norm and quantization sensitivity (Hariri et al., 2025). Our work does not claim that K/V asymmetry is new; it tests and optimizes that asymmetry under speculative acceptance, where a single changed proposal can truncate an entire draft block.

QuantSpec combines self-speculative decoding with a hierarchical 4-bit KV cache (Tiwari et al., 2025). QSpec combines complementary model quantization schemes and reuses model state between drafting and verification (Zhao et al., 2025). In contrast, we study standard two-model speculative decoding and allocate draft-cache precision by measured acceptance sensitivity.

7 LIMITATIONS AND NEXT EXPERIMENTS

The present quantizer uses per-vector symmetric fake quantization. It does not yet implement packed storage, residual windows, per-channel key grouping, or fused attention kernels; therefore analytical cache bytes, not wall-clock latency, are the reliable systems metric. The preliminary results use one model pair and one text distribution. The launched campaign expands to six target–draft pairs, but the final paper also requires multiple seeds, held-out C4/PG19 or LongBench-style contexts, and task-conditioned prompts for code and mathematics.

The additive allocation in Equation 5 ignores interactions between compressed components. We plan to fit a small risk predictor over joint allocations and compare it against uniform precision, KIVI-style quantization, KV-AdaQuant, QuantSpec, and random allocations at matched memory. A packed-cache kernel in vLLM or SGLang is required before making end-to-end throughput and serving-capacity claims.

8 CONCLUSION

Draft KV compression should be evaluated by the speculative work it preserves, not reconstruction error alone. Preliminary results show a large and actionable asymmetry: keys are acceptance-critical routing state, while values tolerate more aggressive compression. This makes K8V4 a stronger operating point than K4V8 at identical memory and motivates acceptance-aware layer-wise allocation. The cross-family campaign will determine whether this behavior is a robust property of speculative decoding or specific to the initial Qwen pair.

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